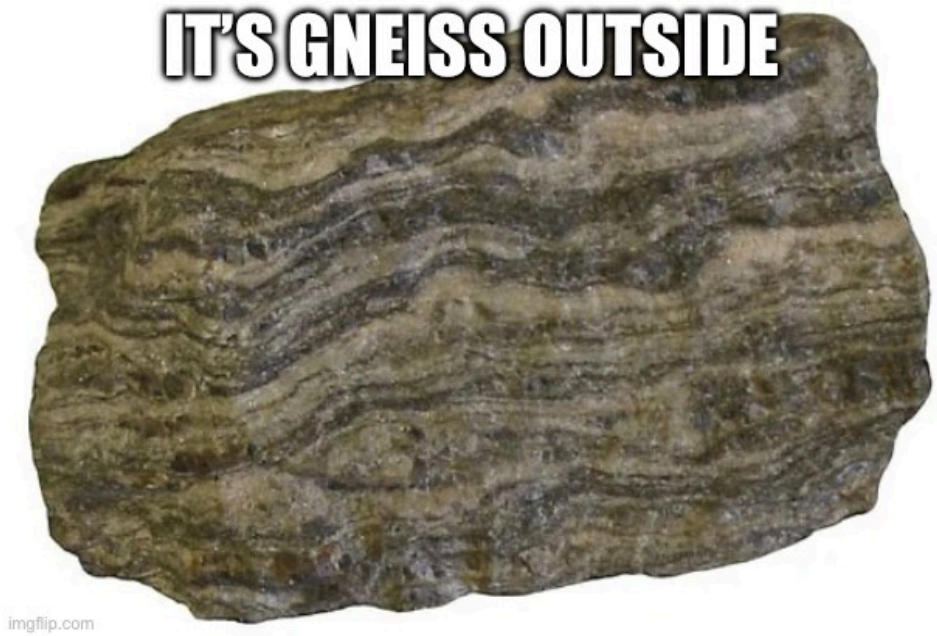
Rocks and Minerals (KEY)

CSE 2025

Tesla STEM High School



Total points: 167

Instructions: Each station is 3 minutes and 20 seconds long. Fill in the tables (each box is worth 1 point unless specified otherwise).

Station 1

Consider Bowen's Reaction Series.	
1) Which mineral crystallizes first?	Olivine
2) In the discontinuous series, which mineral crystallizes just before biotite?	Amphibole
3) Which mineral is most resistant to chemical weathering?	Quartz
4) True or false: Minerals that are higher in the series are more stable at Earth's surface than those that crystallize lower in the series.	False
5) Does calcium rich or sodium rich plagioclase have more silica?	Sodium rich
6) Based on the reaction series, which 3 minerals would you expect to find in gabbro? (1.5 points)	Olivine, Pyroxene, calcium-rich plagioclase (0.5 each)
7) Which branch does amphibole belong to?	Discontinuous branch
8) What mineral forms just before quartz?	Muscovite or white mica
9) If a thick, basalt magma intrusion occurs, a layer of what mineral could you expect to find at the very bottom?	Olivine
10) Can a mafic magma evolve from a felsic one?	No

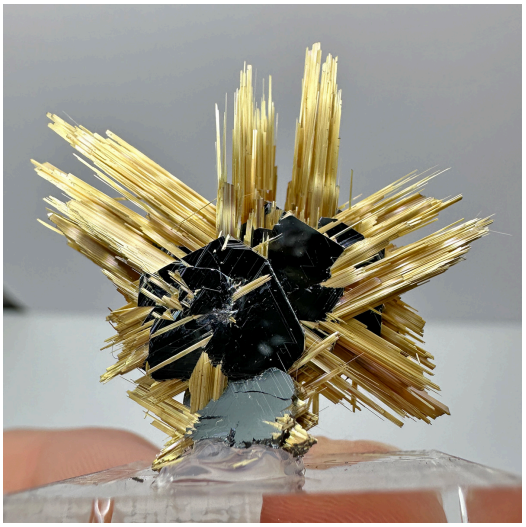
Station 2

11) Identify this mineral.	Dolomite
12) What is its chemical formula?	CaMg(CO3)2
13) What color is its streak?	White
14) Which rock is it most commonly found in?	Dolostone
15) Who is it named after?	Déodat Gratet de Dolomieu (accept Dolomieu)
16) In which era's rock beds is this mineral rare?	Cenozoic
17) What depositional environment is this formed in?	Shallow and warm marine environments
18) Which element in water can turn calcite into this mineral?	Magnesium
19) What is the process described in the previous question called?	Dolomitization
20) What type of bacteria living in what conditions can precipitate this mineral? (2 points)	Sulfate reducing bacteria (1 point) in anoxic conditions (1 point)

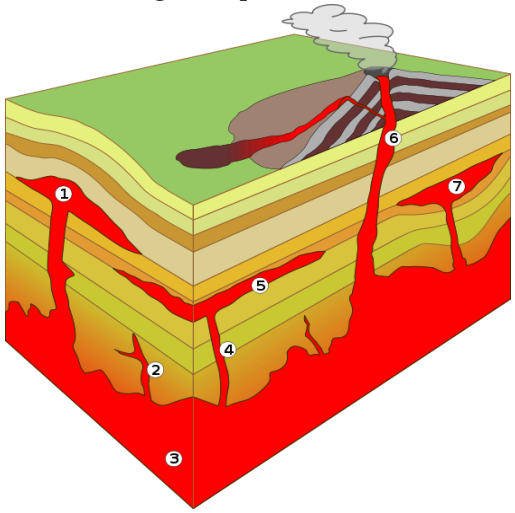
Station 3

21) Identify this mineral.	Gold
22) What is its chemical formula?	Au
23) What color is its streak?	Yellow or gold
24) What is its fracture?	Hackly or Jagged
25) What is this mineral soluble in?	Aqua regia
26) What is its specific gravity?	19.3
27) What type of ore contains large amounts of this mineral, a small amount of quartz, and a lot of black sand?	Blue clay gold ore
28) In its natural mineral form, this mineral almost always has traces of (this mineral is also in the SciOly list):	Silver
29) If added to food, can this mineral increase the nutritional content of food?	No
30) What type of mining is used to extract this mineral from alluvial deposits?	Placer mining

Station 4

	
31) Identify this mineral.	Rutile
32) What is its chemical formula?	TiO_2
33) What is the color of its streak?	Red
34) Small needles of this mineral in gemstones are responsible for what optical phenomenon?	Asterism
35) This mineral is a source of what pigment? What color is this pigment? (2 points)	Titanium Dioxide pigment (1 point) Color: white (1 point)
36) What type of radiometric dating is this mineral commonly used for?	U-Pb dating
37) Why can the mineral be used for the type of dating in the last question?	Uranium can be incorporated into the chemical structure of rutile because of its similar ionic radius and charge to titanium.
38) This mineral is an accessory mineral. What does this mean?	A constituent mineral present in a small quantity and not taken into account when identifying a rock.
39) What are two polymorphs of this mineral?	Anatase, brookite, akaogiite (accept any 2, 0.5 each)
40) What elements, if present, can make this mineral appear black? (2 points)	Niobium and Tantalum

Station 5

Use this image for questions 41-44:  	
41) What is the igneous formation labeled by 1 in the diagram?	Laccolith
42) What is the igneous formation labeled by 2 in the diagram?	Dike
43) What is the igneous formation labeled by 5 in the diagram?	Sill
44) What is the igneous formation labeled by 3 in the diagram?	Batholith
45) Is ice a mineral?	Yes
46) Explain your answer above. (2 points)	Ice meets all the requirements for being a mineral (1 point) It's naturally occurring, inorganic, solid, has a definite chemical composition, and an ordered internal crystalline structure (1 point)
47) Which metamorphic facies has the lowest metamorphic grade?	Zeolite facies
48) Which metamorphic facies occurs at low temperature and high pressure?	Blueschist facies
49) What is the state mineral of Illinois?	Flourite
50) What is Nevada's state rock?	Sandstone

Station 6

51) Identify this mineral.	Vanadinite
52) What is its chemical formula?	$Pb_5(VO_4)_3Cl$
53) What color is its streak?	Pale or Brownish yellow
54) How does this mineral form?	The alteration and oxidation of primary lead ore deposits in the presence of vanadium-rich solutions.
55) What variety of this mineral contains a high amount of arsenic impurity?	Endlichite
56) If the arsenic substitution continues, what will the mineral become?	Mimetite
57) What two elements are extracted from this mineral?	Vanadium and lead (0.5 each)
58) Which country was this mineral first discovered in? What was it originally named?	Mexico, brown lead (0.5 each)
59) This mineral is commonly associated with the oxidation of what mineral?	Galena
60) What country has the best specimen of this mineral?	Morocco

Station 7

61) Identify this mineral.	Galena
62) What is its chemical formula?	PbS
63) What color is its streak?	Lead gray/Dark gray/Black (accept any)
64) What state(s) is this the state mineral of (state all states applicable): (1.5 points)	Kansas, Wisconsin, and Missouri (0.5 points each)
65) What metal is this a primary ore for?	Lead
66) What are the most common weathering products of this mineral?	Cerussite and Anglesite (0.5 points each)
67) What is the variety of this mineral that contains silver? Where is it primarily found? (2 points)	Argentiferous galena (1 point), found in hydrothermal veins (1 point)
68) What is its crystal system?	Cubic
69) The pollution of which heavy metals are associated with mining of this mineral?	Lead and Cadmium (0.5 points each)
70) A variety of this mineral containing Bismuth will have what cleavage?	Octahedral

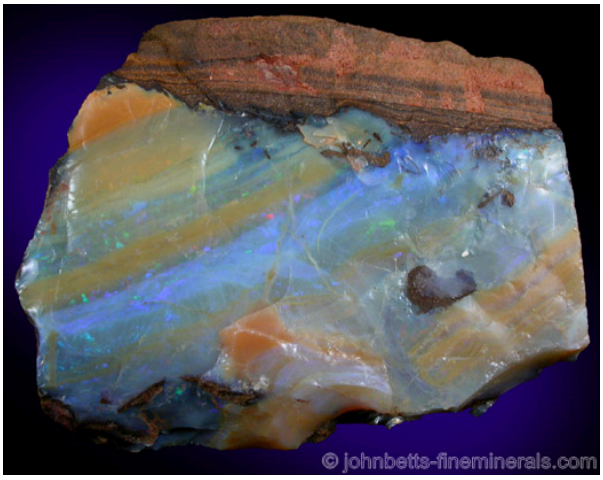
Station 8

71) Identify this mineral.	Olivine
72) What is its chemical formula?	(Mg,Fe)2SiO4
73) What is its crystal system?	Orthorhombic
74) Is this mineral commonly found in sedimentary rocks? Why or why not? (2 points)	No (1 point), because of its rapid weathering (1 point)
75) This mineral is a nesosilicate. Describe the structure of nesosilicates.	Isolated silicon-oxygen tetrahedra connected by interstitial cations.
76) What are the two endmembers of the solution series that this mineral is in?	Forsterite and Fayalite (0.5 each)
77) Which endmember is more common?	Forsterite
78) What happens to this mineral at a depth of 410 km?	It transitions into wadsleyite.
79) This mineral is the most abundant mineral in what part of the Earth?	Upper mantle
80) In the image, what is the rock that the mineral is embedded in?	Basalt

Station 9

81) Identify this mineral.	Talc
82) What is its chemical formula?	$Mg_3Si_4O_{10}(OH)_2$
83) What is its crystal system?	Monoclinic
84) What type of plate boundary is this mineral typically found in?	Convergent plate boundaries
85) How did most of the deposits of this mineral in the US form?	Heated waters carrying magnesium and silica reacted with dolomitic marbles.
86) Why is this mineral so soft?	The thin silicate sheets are held together by weak van der Waals forces.
87) What metamorphic rock is primarily composed of this mineral?	Soapstone
88) What is this mineral's tenacity?	Sectile
89) Which form of this mineral is rarer than other forms?	Crystallized form
90) Is this mineral used as a conductor or an insulator?	Insulator

Station 10

	
91) Identify this mineraloid.	Opal
92) What is its chemical formula?	$SiO_2 \cdot nH_2O$
93) What color is its streak?	White
94) What is this mineraloid composed of?	Tiny spheres of silica
95) Why can some specimen refract light while others can't?	The density and pattern of aligned silica spheres can reflect light. If the spheres are too disordered or compact, the specimen cannot reflect light.
96) Can this mineraloid be scratched by a steel nail?	Yes
97) What type of solution does this mineraloid form from?	Silica-rich solutions
98) How does crazing affect some of these mineraloids?	Causes them to form internal and external cracks.
99) What are schillers?	Color flashes
100) What is the range of water content in this mineraloid?	3-21%

Station 11

101) Identify this mineral.	Zircon
102) What is its chemical formula?	$ZrSiO_4$
103) What is its crystal system?	Tetragonal
104) What is it an ore for?	Zirconium
105) Can this mineral be scratched by glass?	No
106) What is metamictization?	The mineral's crystal structure is damaged due to radioactive decay
107) How is the color of this mineral changed by humans?	Heat treatment
108) What luster does this mineral have when it is radioactive?	Pitchy
109) A darker brown or black color is due to what impurity?	Iron oxides
110) What is this mineral's tenacity?	Brittle

Station 12

111) Identify this rock.	Pegmatite
112) What is its texture?	Very coarse grained/pegmatitic
113) Is this rock intrusive or extrusive?	Intrusive
114) Are mafic versions of this rock common?	No
115) Lithium is extracted from specimens containing what mineral?	Spodumene
116) How do bodies of this rock compare to bodies of other intrusive rocks?	Pegmatite bodies are smaller
117) In a magma body, is this rock the first or last to crystallize?	Last
118) What rock's composition is most similar to this rock's composition?	Granite
119) This rock forms from magmas rich in what?	Water
120) What is the role of the compound you identified in the previous question in promoting the growth of large crystals?	Lowers viscosity and enhances ion mobility

Station 13

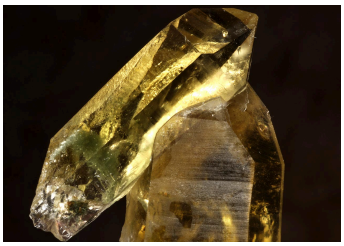
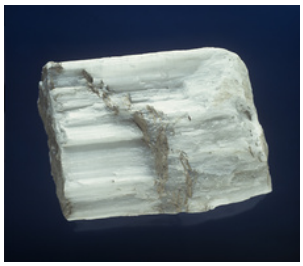
121) Identify this rock.	Conglomerate
122) What is its texture?	Clastic
123) In what depositional environment are the grains well rounded and poorly sorted?	Fluvial
124) In what depositional environment are the grains well rounded and poorly sorted?	Deepwater marine
125) What is the matrix?	The fine-grained material that fills the spaces between the clasts and cements them together
126) What does the rounding of clasts indicate about their history?	They were transported large distances from their original source or they were in high energy environments
127) When the grains are poorly sorted, what does this indicate about the past environment?	There was variable energy or rapid deposition
128) What does it mean when this rock is clast supported?	The clasts touch each other and the matrix fills the voids
129) What does it mean when this rock is matrix supported?	The clasts are not in contact and the matrix surrounds each clast
130) Which rock is similar to this rock but has angular fragments?	Breccia

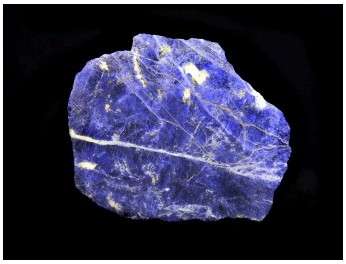
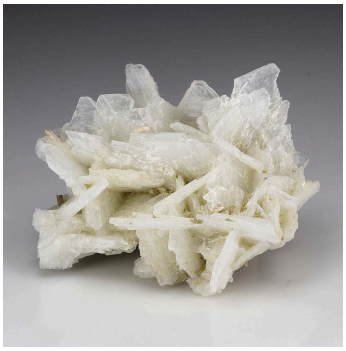
Station 14

131) Identify this rock.	Amphibolite
132) What is its texture?	Non foliated/weakly foilated
133) What is its parent rock?	Basalt/mafic igneous rocks
134) What type of metamorphism usually forms this rock?	Regional metamorphism
135) What temperatures does this rock usually form under?	500-750 degrees C
136) What pressures does this rock usually form under?	7-8 kbar
137) Which minerals is this rock primarily composed of?	Amphibole (hornblende and actinolite), and plagioclase feldspar
138) In the image, which mineral makes up the black portions?	Hornblende
139) Which plate boundary is this rock associated with?	Convergent plate boundaries
140) What are high quality specimen used for?	Architectural use (floor tiles, panels, countertops etc.)

Station 15

Identify each rock/mineral.

			
141) Malachite	142) Citrine	143) Alabaster	144) Scoria
145) Phyllite	146) Tremolite	147) Chalcedony	148) Arkose
149) Hornblende	150) Willemite	151) Hematite	152) Spodumene

			
153) Epidote	154) Sodalite	155) Barite	156) Beryl
			
157) Albite	158) Fluorite	159) Rose Quartz	160) Coquina

Captain’s Summer Exchange - Tesla STEM High School - Repaired practice edition - Answer key

Division C | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

27. (6 points) Authored replacement for source item Question 27: Solve this case and explain your reasoning: An unknown scratches gypsum but is scratched by fluorite. What can you conclude?

reasoning: Gypsum is Mohs 2; fluorite is Mohs 4.
The test brackets hardness between about 2 and 4.
Calcite at 3 is compatible, but additional evidence is needed. (3 points).

conclusion: A bracket is evidence, not a unique identification. (3 points).

60. (6 points) Authored replacement for source item Question 60: Solve this case and explain your reasoning: A dark specimen has reddish-brown streak and weak magnetism. Which iron mineral is favored?

reasoning: Compare hematite and magnetite rather than using color.
Hematite commonly gives reddish-brown streak; magnetite is strongly magnetic.
Favor hematite while checking other properties. (3 points).

conclusion: Use diagnostic combinations. (3 points).

70. (4 points) Authored replacement for source item Question 70: Differentiate mineral and rock cleavage.

Unit: Metamorphism, stress, and mineral assemblages.
Worked-case data: A carbonate-rich rock near an intrusion recrystallizes without melting. Propose an interpretation.

answer: Mineral cleavage is crystal breakage; rock cleavage is an aggregate planar fabric. (4 points).

95. (6 points) Authored replacement for source item Question 95: Solve this case and explain your reasoning: A massive specimen lacks visible faces. Does it lack crystalline order?

reasoning: Habit describes external appearance.
Many small intergrown crystals can produce a massive appearance.
Other structural and physical observations are needed. (3 points).

conclusion: Absent faces do not establish an amorphous material. (3 points).

96. (6 points) Authored replacement for source item Question 96: Solve this case and explain your reasoning: Why does a fully shared silica framework have SiO_2 rather than SiO_4?

reasoning: Each tetrahedron has four oxygen corners.
Each corner is shared by two tetrahedra.
Count four halves = two oxygen atoms per silicon. (3 points).

conclusion: Sharing changes atom bookkeeping. (3 points).

123. (4 points) Authored replacement for source item Question 123: What is a pseudomorph?

Unit: Formation environments, crystal systems, and habit.
Worked-case data: A massive specimen lacks visible faces. Does it lack crystalline order?

answer: A replacement mineral preserves the external form of an earlier material. (4 points).

124. (4 points) Authored replacement for source item Question 124: What do vesicles record?

Unit: Igneous processes, textures, and Bowen’s series.
Worked-case data: Granite and rhyolite have similar broad composition but different textures. Explain.

answer: Gas bubbles trapped during solidification, without uniquely fixing composition. (4 points).